Design Guide for Saudi Arabia

Minimum design standards for electrical services



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Design Guide for the Kingdom of Saudi Arabia:

Minimum design standards for electrical services

DGSA02: 2021



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ISBN 978-1-914543-01-2 (PDF)

Registered charity number 278104

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Editing, typesetting, layout and design by CIBSE Knowledge

Cover illustration: Saudi Arabia skyline silhouette ©Shutterstock/Sky and glass

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Note from the publisher

This publication is primarily intended to provide guidance to those responsible for the design, installation, commissioning, operation and maintenance of building services. It is not intended to be exhaustive or definitive and it will be necessary for users of guidance to exercise their own professional judgement when deciding whether to abide by or depart from it.

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This document aims at establishing the minimum design standards for the design of electrical systems for buildings in the Kingdom of Saudi Arabia. It does not relieve the designer of responsibility for accurately determining design arrangements and/or complying with the local authority requirements and/or relevant international codes or regulatory requirements. It should be used for initial design guidance, and provides minimum design requirements for specific design issues.

This guide has been produced in good faith by the CIBSE UAE Technical Committee to promote best engineering practices in the Kingdom of Saudi Arabia, without legal or financial liability of CIBSE UAE or the authors, in order to assist engineers coming from elsewhere in the world to work in the Kingdom of Saudi Arabia. It should be read alongside existing CIBSE guidance relevant to the region, including *Buildings for Extreme Environments: Arid*.

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1 Design codes, standards and guidance

The design, specification and installation of electrical services should be in accordance with good engineering practice and conform to, but not be limited to, the following standards, codes of practice and regulations applicable in the Kingdom of Saudi Arabia.

Table 1.1 Codes, standards and guidance

Source/reference	Code, regulation, guideline reference
Saudi Building Code National Committee	SBC-201: *Saudi Building Code: General* (SBCNC, 2018a) SBC-401: *Saudi Electrical Code* (SBCNC, 2018b) SBC-801: *Saudi Fire Protection Code* (SBCNC, 2018c) SBC-1001: Saudi Green Building Code (SBCNC, 2018)
Saudi Electricity Company	DPS: Distribution Planning Standards* DES: Distribution Engineering Standards SDMS: SEC Distribution Material Specifications* SDCS: SEC Distribution Construction Standards* Various circulars (https://www.se.com.sa)
Saudi Standards, Metrology and Quality Organization (SASO)	Various standards. The latest version of SASO standards are applied. In case of no relevant SASO Standards, the latest version of IEC and other standards shall be applied.
Electricity and Cogeneration Regulatory Authority (ECRA)	*Electricity Service Guide Saudi Arabia* (ECRA, 2019) *The Saudi Arabian Grid Code* (ECRA, 2020) *The Saudi Arabian Distribution Code* (ECRA, 2018)
Ministry of Water and Electricity	Residential and commercial buildings electrical installation guidance for 400/230 voltage system
Mostadam	Comprehensive sustainability rating and certification system for KSA (https://www.mostadam.sa)
International Electrotechnical Commission (IEC)	Various standards
British Standards Institution (BSI)	BS 7671: *Requirements for Electrical Installations. The 18th Edition IET of the Wiring Regulations* (BSI, 2018/2020)
National Fire Protection Association	NFPA 70: *National Electrical Code* (NFPA, 2020a)
	NFPA 72: *National Fire Alarm and Signalling Code* (NFPA, 2019a)
	NFPA 101: *Life Safety Code* (NFPA, 2021)
	NFPA 110: *Standard for Emergency and Standby Power Systems* (NFPA, 2019b)
American National Standards Institute (ANSI)	
Institute of Electrical and Electronics Engineers (IEEE)	
Society of Lighting and Lighting (SLL)	Various SLL Lighting Guides
Chartered Institution of Building Services Engineers (CIBSE)	Commissioning Code L: *Lighting* (CIBSE, 2018)
International Commission on Illumination (CIE)	

* Details for these publications are given in Appendix 1

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In general, it is recommended that designers consider the following electrical design elements and incorporate as appropriate:

- electrical infrastructure
- substation design
- medium voltage (MV) distribution
- low voltage (LV) distribution
- small power design
- power supplies for mechanical, ELV systems, vertical transportations, security, furniture, fixtures, equipment, etc.
- power factor correction
- harmonic suppression
- electrical metering
- resilience and emergency power generation
- electrical vehicle charging
- life safety systems
- uninterrupted power supply (UPS)
- internal and external lighting
- façade lighting systems
- lighting control systems
- aviation obstruction lighting
- emergency lighting
- fire alarm and detection
- fire telephone and refuge systems
- alarms for disabled persons
- earthing and bonding
- lightning protection and transient over-voltage protection
- cabling and cable containment and support systems.

2 Incoming services

The designer is advised to contact and make applications to the Saudi Electric Company (SEC) throughout design stages to secure the required electrical connections to the site.

Connections should be established at low voltage (400/230 V) or medium voltage (13.8 kV or 33 kV) as required to suit the site load requirements.

In the case of a medium voltage (MV) supply, the designer should include MV switchgear/ring main unit (RMU), transformers and associated rooms as required to allow internal low/medium voltage (LV/MV) distribution within the site.

All incoming supplies, whether low/medium voltage, should be established within a suitably sized dedicated electrical rooms, suitably arranged with respect to incoming service point, site distribution and ease of access, maintenance and plant replacement in accordance with SEC requirements.

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Electrical supply characteristics at source (incoming low service, secondary side of transformer for medium voltage supply) should generally be considered as follows during initial design development:

— *primary voltage*: 13.8 kV or 33 kV, 60 Hz, 3-phase, 3-wire

— *secondary voltage*: 400/230 V, ±6%, 60 Hz, 3-phase, 4-wire, solidly earthed

— *fault current*: 40 kA for two seconds at main distribution board when transformer rating is 1500 kVA and 25 kA for one second

— *power factor correction*: greater than 0.95

— *earthing arrangement*: TN-S

— *voltage drop*: see Table 2.1.

Table 2.1 Voltage drops

Installation type	Lighting (%)	Other uses (%)
Low voltage installation supplied from private LV supply	6	8
Low voltage supplied directly from a public low voltage distribution system	3	5

Note: For voltage drops in consumers' installations, SBC-401, clause 52-5 (SBCNC, 2018b) recommends that, in practice, the voltage drop between the origin of consumer's installation and the equipment should not be greater than 4% of the nominal voltage of the installation.

The designer will need to determine final supply characteristics/local requirements with the SEC during the design process based on six regions (Eastern, Central, Northern, Northwest, Midwest, and southwest) and the thirteen provinces (located within those six regions).

3 Electrical loads and diversity

Electrical loads should be assessed based on individual site requirements considering the local method of load assessment, e.g. load per point for socket outlets and lighting points in accordance with Saudi Electrical Company (SEC) DPS regulations and the actual project requirements.

While the local means of load calculation should be used to establish the total connected load (TCL) requirement of each site, designers should endeavour to incorporate design solutions that minimise the operational loads of buildings to reduce overall energy consumption, considering the demand factors and coincident factors listed in SEC Distribution Planning Standard DPS-01: *Estimation of customer load guideline**, 'Connected load and demand load'.

Lighting should particularly be targeted in this respect, with designers required to restrict lighting power densities to the levels defined within ASHRAE Standard 90.1 (ANSI/ASHRAE, 2019).

Diversity should be applied based on the application considering likely usage, together with local norms. The designer should agree all applied diversities with SEC as part of the design process. The maximum demand permitted on the main distribution board should be in accordance with SEC regulations.

Note: Within KSA guidance, diversity is often defined as coincidence factor.

* See Appendix 1

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4 Main distribution

4.1 General

All distribution equipment (main, sub and final) should be arranged in accordance with SEC regulations and requirements, with particular consideration given to location, access, equipment clearances, distribution routes and lengths, avoidance of any wet services in the locality (i.e. above, adjacent, within area), presence of other services etc., as required. Designers should ensure all required design approvals are obtained as part of the design process.

4.2 Equipment specifications

In general, the electrical panel specifications shown in Table 4.1 should be considered as a minimum.

Table 4.1 Minimum electrical panel specifications

Type	IP rating	Minimum form of construction	Reference (see Appendix 1)
Main distribution boards	IP43	Form 4b, type 6	31-SDMS-05 31-SDMS-01
Motor control centres	IP43	Form 2b, type 2	31-SDMS-05
Sub-main distribution boards (SMDB)	IP43	Form 2b, type 2	31-SDMS-05

Note: Table 4.1 is based on internal installation only and guidance only as this is not clearly defined within the referenced documentation. Where switchgear is installed externally it should have a minimum rating of IP65 and suitably rated for ambient conditions in line with standards of good practice*

All outdoor equipment enclosures/material shall be weatherproof and all metal parts shall be corrosion and/or abrasion resistant, and the degree or protection shall be IP54 as per IEC 60529 (IEC, 1989/1999/2013) or equivalent National Electrical Manufacturers Association (NEMA) Standard. The degree of protection for all indoor equipment shall be IP41 as per IEC 60529 or equivalent NEMA Standard.

The designer should consider requirements for individual projects, and areas of projects, and tailor the specifications where appropriate. This could be defined by project type (e.g. form ratings should be increased where power resilience is required and supply isolations will cause significant disruption to the building or area functions, such as data centres or healthcare projects), equipment location (equipment installed externally or in a plant room, for instance) or by individual project or end user requirements.

All switchgear should be factory-assembled, either by the protective device manufacturer or by a panel builder licensed by the protective device manufacturer and approved by SASO/SEC. All switchgear should be provided with ammeters, indication lamps, voltmeters (reading phase-to-neutral and phase-to-phase) as applicable, and with provisions for the installation of metering equipment required by SEC.

As a minimum, fault currents of all electrical equipment and ancillary devices should be rated in accordance with SEC standard requirements. Designers should, however, conduct full system calculations for each project to determine potential fault currents throughout all designed systems. Should the fault current in any area of an installation be above normal SEC regulations, the designer should increase the fault ratings of the associated equipment accordingly.

It is recommended that as a minimum, the spare equipment capacity allowances shown in Table 4.2 should be considered as part of the electrical distribution design.

* As per 01-SDMS-01, Rev. 01

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Table 4.2 Recommended future spare capacity for electrical equipment

Equipment	Typical spare way allowance
Main LV switch panels	20%
Sub-main distribution boards	10% or two TP ways, whichever is greater
Motor control centres	10% or two TP ways, whichever is greater
General final circuit distribution boards	20% or two TP ways, whichever is greater

Additional spares are to be considered on a project-by-project basis with appropriate allowances made considering project type, equipment location and any potential future project expansions planned.

It is important to note that equipment spares allowances relate to physical electrical equipment only (i.e. space provision), and should not be confused with overall building load spare allowances — these should be applied according to project-specific requirements.

4.3 Power factor correction

From May 2022 as per ECRA letter ref. 02/27/33, designers should make provision for the installation of power factor correction to achieve a power factor of between 0.95 (minimum) and unity (maximum). The capacitors to be used at the terminals of inductive devices such as motors and transformers or where the level of load is reasonably constant, shall be fixed type to form a continuous level of compensation. Control shall be either manual or semi-automatic. Where the active power or reactive power variations are relatively large or the kVAr rating of the capacitors is more than 15% of the supply transformer, automatic capacitor banks shall be used to provide automatic control of compensation maintaining the power factor within close limits around a selected level.

With reference to SEC Distribution Material Specification 43-SDMS-03: *Power factor correction for low voltage power systems**, designers should ensure that the capacitor sizes used in the banks are arranged to provide suitable stages to maintain the operational power factors. Typically, 25 kVAr stages should be considered as a maximum. However, smaller stages should be adopted where required to suit installation location and specifics. Designers should offer consideration to detuned capacitor banks where required to combat the effects of installation harmonics.

In addition to correcting the power factor of installations at supply intake points, designers should consider localised downstream correction methods where required and should ensure that all highly inductive loads are provided with integral means of correction.

All power factor correction banks should be accommodated in a suitable prefabricated cabinet or in a series of modular cubicles assembled to form a composite unit. Correction banks should consist of control relays, switching equipment, protective devices and means of local isolation to control the automatic switching of capacitors in response to changes in load power factor.

HRC fuses or MCCBs shall be used for short-circuit protection. The connecting cables and circuitry shall be selected so that 1.3 times the rated capacitor current can be permanently handled.

The maximum admissible over-current of 1.3 × IR (rated current) to IEC 60831-1 (IEC, 2014) shall be maintained by all capacitors. Thermal/thermal magnetic or electronic over-current relays shall be used for over-current protection. HRC fuses do not protect against overload.

The HRC fuse rating should be 1.6 to 1.8 times nominal capacitor current. HRC fuses shall not be used for switching because of the risk of arcing.

* See Appendix 1

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The designer should ensure that all equipment is suitable for operation from the supply voltage and frequency, including the tolerances specified in the incoming services section and the operational temperatures defined previously.

4.4 Surge protection

Surge protection devices should be considered to offer protection to electrical and electronic components from the secondary effects of transient overvoltage.

Designers should arrange surge protection in accordance with the recommendations of IEC/BS EN 62305 (BSI, 2011–2012). As a minimum, protection should generally be considered on all copper power, data communication, signal and telephone lines where they enter or leave a building to prevent transient damage.

4.5 Harmonic suppression and filtration

Designers should assess the likely harmonic components of each project and, where required, implement suppression or filtration equipment to reduce harmonic distortion and/or space to accommodate future suppression-filtration equipment if deemed necessary at a later date.

Harmonic filtration equipment should typically be provided at main distribution boards, and filtration should also be provided locally, or integral to equipment that has the potential to create harmonics within installations, e.g. variable speed drives, UPS systems etc.

4.6 Metering

With reference to SEC Distribution Material Specifications* 38-SDMS-03, 40-SDMS-01, 40-SDMS-02A, 40-SDMS-02B, 42SDMS-01, 42-SDMS-03, 32-SDMS-02, 32-SDMS-05 and 32-SDMS-06, utility metering should be provided at all supply intake points in accordance with SEC regulations.

In addition to the mandatory utility metering requirements, landlord sub-metering should be considered within installations to monitor and record energy consumption within installations and assist with potential energy savings. Landlord sub-metering should typically be provided to the following as a minimum:

— all standalone buildings on site

— all third-party tenanted spaces

— specific areas identified by the client, end user or otherwise defined in project request for proposal (RFP).

In addition to the above, consideration should also be given to the following on a project-by-project basis:

— sub-main distribution boards

— motor control centres

— departmental segregations.

In general, designers should ensure that all landlords' sub-meters are digital, complete with data-logging capabilities and links to a centralised system (either via a building management system or standalone system) to allow centralised monitoring and recording.

* See Appendix 1

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5 Standby generation and distribution

At the start of each project, designers should assess the need for standby generation and distribution to provide back-up electrical power supplies as necessary. Requirements for standby power should be dependent on:

— project type and criticality
— project size and configuration
— project location and reliability of local supplies
— fire and life safety requirements
— Saudi Civil Defence requirements
— SEC regulations.

Standby generators should typically be provided to offer redundancy and resilience to the normal utility electrical power supplies and should serve all fire and life safety equipment and services and any specific equipment deemed essential for the function of the project.

The following lists provide a high-level summary of the type of equipment and services to be considered. The list is not exhaustive, and designers should offer specific consideration to individual project requirements.

— ***Statutory life safety systems***:
  - Emergency escape lighting*
  - Standby lighting
  - Exit and directional escape signage
  - Fire alarm system
  - Fire network pumps
  - Smoke ventilation and extract fans
  - Public address/voice alarm (PA/VA) systems
  - Fireman's lifts.

— ***Essential power systems***:
  - Data, GSM and telecom server equipment
  - BMS and security equipment, except distribution racks, doors, barriers, CCTV cameras, etc.
  - HVAC equipment serving data and telecom server room
  - Selected lifts (as advised by fire and life safety (FLS) and vertical transportation (VT) teams)
  - Sump pumps, booster pumps, circulation pumps, sewage and stormwater lifting pumps
  - Coldstores/freezers
  - Cinemas
  - Uninterruptible power systems (UPS)

— ***Essential lighting***:
  - Electrical equipment rooms
  - Fire command centres

* This generally considers escape route lighting, open area anti-panic lighting for areas greater than 60 m^2, high-risk task areas and toilets with areas greater than 8 m^2.

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- Fire pump rooms
- Generator rooms
- Interior/exterior stairways and ramps
- Discharge vestibules
- Exterior landings for exit doorways
- Public restrooms (greater than 28 m^2) and in all toilets for the disabled emergency lighting

Electrical distribution of standby generator-backed power supplies should be arranged in accordance with the requirements of Saudi Civil Defence, and NFPA 70: *National Electrical Code* (NFPA, 2020a), with appropriate segregation provided between emergency, legally required and optional standby systems in accordance with the aforementioned code.

Designers should consider requirements for fire-rated cabling and dual diverse supplies for all generator-backed supplies to offer appropriate levels of supply redundancy and resilience depending on equipment or service type, function, location and associated code requirements.

Designers shall take into consideration all requirements of the civil defence authority (CDA) regarding the locations of the fuel bulk tank and the minimum required capacity of the tank that is needed for standby power system to cater all life safety equipment.

Automatic transfer switches (ATS) should be provided to transfer loads over to the essential generator-backed supply in the event of any outages to the normal utility mains supply. ATSs, together with associated standby generation equipment and controls, should be arranged so that all loads are transferred and available in accordance with the maximum time periods prescribed in NFPA 70: *National Electrical Code* (NFPA, 2020a) for each standby system category (i.e. emergency, legally-required and optional).

Where standby generation plant is provided appropriate consideration should be given to the generator requirements, with specific consideration given to generator:

— installation
— maintenance access and replacement
— testing
— spare requirements
— de-rating factors, i.e. due to air density (altitude), humidity, ambient temperatures
— operational requirements
— ventilation
— fume exhaust
— fuel storage and refuelling
— acoustic requirements
— vibration.

The designer should consider the above, together with any other project or generator specifics, to ensure that the design solutions ultimately satisfy the project requirements.

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6 Uninterruptible power supplies

Uninterruptible power supplies (UPS) should be provided on a project-by-project basis depending on:

— project type

— services installed

— criticality of service continuity

— end user requirements.

UPS systems should typically be utilised to serve critical site services, such as data and information technology equipment, and critical site functions (e.g. data centre, server room, security systems, selected healthcare areas or departments, critical laboratories etc.) This should provide a continuous, clean, stable power supply (i.e. free from spikes, surges, harmonics etc.) irrespective of mains fluctuations or interruptions.

The location of UPS units within installations (i.e. centralised at source, semi-centralised at distribution branch or project department, or localised to specific equipment or services) should be determined with due consideration to the individual project requirements, and assessed based on utilisation, resilience, economy and maintenance.

The type of UPS selected should be dependent on usage, load, level of criticality, autonomy, space requirements and project budget. In general, static UPS systems should be used in most applications. However, consideration should also be given to rotary UPS systems where appropriate to satisfy the needs of specific projects. Designers should consider UPS autonomy requirements and redundancy configurations for each application and present to the end user for review and approval as part of the design process.

7 Cabling

Designers should select cable types and installation methods to suit individual applications; however, the cable types shown in Table 7.1 should typically be considered.

Table 7.1 Cable types to be considered

System	Typical cable type
Incoming MV supply	As per SEC (DPS-09, DPS-10 and 11-SDMS-03)
	For 13.8 kV: • 3 × 300 mm^2 Cu/XLPE/PVC; 390 A; max. firm capacity 9 MVA • 3 × 185 mm^2 Cu/XLPE/PVC; 290 A; max. firm capacity 7 MVA • 3 × 500 mm^2 Al/XLPE/PVC; 320 A; max. firm capacity 7.6 MVA • 3 × 300 mm^2 Al/XLPE/PVC; 300 A; max. firm capacity 7 MVA
	For 33 kV: • 3 × 240 mm^2 Cu/XLPE/PVC; 350 A; max. firm capacity 20 MVA • 3 × 185 mm^2 Cu/XLPE/PVC; 290 A; max. firm capacity 17 MVA
Transformer to main LV panel	Multiple single core XLPE/PVC parallel feeds (as per DPS-02, rev. 02)
Sub-mains distribution	Multi-core armoured cabling (Cu/XLPE/SWA/LSF) and/or busbar distribution (typically used for high-rise developments and industrial applications)
Essential sub-mains distribution	Fire-rated multi-core armoured cabling and/or fire-rated busbar distribution systems
Final circuit cabling	Typically, single core Cu/LSF cabling and/or prefabricated wiring where applicable and approved
Sub-main and final circuit cabling installed externally	Multi-core armoured cabling (Cu/XLPE/SWA/LSF)

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Final cable selections should be undertaken with due consideration to project and supply type, installation routes and methods, containment and support system types, and the requirements of SEC and the end user.

Designers should complete full cable calculations as part of the design to establish minimum cable sizes for each circuit in accordance with the aforementioned requirements. Calculations should consider all installation specifics, including but not necessarily limited to:

- circuit type, length, load and supply arrangement (i.e. parallel feeds)
- cable selection and installation method
- protective device selection, size and type
- correction factors for ambient temperature, grouping, thermal installation and operating temperature
- required circuit disconnection time
- earth fault loop impedance and prospective fault current at source and through installation
- voltage drop at source and through installation.
- the current carrying capacity of the cables/feeders, which shall be equal or greater than the rated current of the overcurrent protection device (OCPD) as per IEC 60364-4-43 (IEC, 2008).

All calculations should be undertaken using recognised electrical design software.

7.1 Cable containment and cabling installation

Designers should ensure that all lengths of cabling are suitably supported or contained within dedicated cable containment systems. Support and containment types should be selected according to:

- cable type
- circuit type
- supply point
- installation route
- installation specifics and means of access.

Means of support and containment systems should be designed and installed in accordance with SEC regulations, BS 7671 (BSI, 2018/2020) and the manufacturer's guidelines. These requirements should be followed for selecting appropriate system types, providing all necessary circuit segregation and separation, sizing systems to suit cabling arrangements and providing associated supports and support systems to satisfy cabling requirements and applied installation weights.

Table 7.2 below defines the typical means of cable supports and containment that should generally be considered for installations.

Designers should note that this is provided for reference purposes only, to offer guidance. Designers should select appropriate methods for each project with due consideration to installation specifics and individual project requirements and considerations.

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Table 7.2 cable supports and containment

Electrical system/sub-system	Typical means of installation and/or containment system
MV switchgear to transformer	Typically installed through trench as per SEC standard
Transformer to main LV panel	Typically installed through trench as per SEC standard
Sub-distribution from main LV panel (i.e. to sub-main distribution boards, motor control centres, etc.)	Cable tray, ladder or busbar for main/primary routes Clipped direct for individual/secondary routes
Final circuit distribution (i.e. to power and lighting points)	Cable tray for main/primary routes Conduit or cable trunk for individual/secondary routes
Sub-distribution and final circuit distribution installed externally	Typically installed in cable ducts in hard standing areas; buried direct cabling permitted in soft landscape areas only
Emergency lighting Central battery wiring	Cable tray, trunking or basket for main/primary routes Conduit or clipped direct for final connections
Fire alarm and detection wiring	Cable tray for main/primary routes Conduit, trunking or clipped direct for final connections
Extra-low voltage systems	Cable trunking or basket tray for main/primary routes Conduit or trunking for final connections

All cable containment should be sized in accordance with SBC-401 (SBCNC, 2018b), IEC 61537 (IEC, 2006) and BS 7671 (BSI, 2018/2020), with due consideration given to:

— cable numbers and sizes

— cable spacing allowances based on touching or spaced-cable installation methods

— containment space factors, taking into account containment thicknesses and means of cable installation

— spare capacity allowances.

Spare capacity allowances should be selected considering containment type, usage, installation location, likelihood of future installation extensions or expansions, and end user requirements. A spare capacity of 20% should typically be allowed as a minimum; however, final allowance should be selected with due consideration to the aforementioned criteria. All spare capacities should be applied after cable-spacing factors have been considered.

Containment systems should be arranged to offer appropriate segregation and separation between:

— cabling serving different systems or different classifications

— essential and non-essential supplies or circuits

— supplies from different supply sources or distribution paths.

Segregation and separation should be arranged in accordance with SBC-401 regulations and IEC 61537.

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Designers should consider the arrangement of containment systems and cabling installation in coordination with other services installed and the installation specifics, with sufficient clearance provided on the sides and above containment systems to allow personnel access for cable installation, replacement and upgrading purposes.

Dimensions associated should be dependent on installation specifics (e.g. containment widths, heights, access and usage, cable sizes and quantities, general arrangements, presence of other services, etc.)

However, the following should be applied as a minimum wherever possible:

— side(s) of containment: 600 mm

— above containment: 300 mm.

The application of the above should be considered on a project-by-project basis. All dimensions applied should be coordinated with the general project arrangements and should not compromise other installations or result in uneconomical or unviable service void requirements.

Containment systems should be supported at regular intervals in accordance with the guidelines of the manufacturers and load calculations for support systems based on installation methods (wall mounted, floor mounted or ceiling suspended), width of containment system and number of layers, taking into account the applied weights of installed and potential future cabling of installations. Seismic supports are to be utilised wherever required as per seismic zone classification of the location of the site.

All containment should be selected with due consideration to its installed environment, with appropriate protection provided against corrosive or polluting substances, extremes of temperature and excessive solar radiation as and where required.

8 Small power and power supplies

General small power and fixed power supplies should be arranged to suit the requirements of each installation. Small power outlets should be provided to satisfy the portable appliance and equipment requirements of installations, with due consideration given to:

— installation or area type and usage

— likely installation or area requirements, based on:

  • furniture and equipment layouts

  • potential positions of potable appliances or equipment

— SBC-401 (SBCNC, 2018b) regulations, particularly in terms of load allocation

— end user requirements.

The above requirements should be established throughout the design process so that the final provisions satisfy the likely requirements of each installation.

The following provides an indicative description of the typical minimum requirements to be considered for generic installation areas. Designers should, however, consider each project on a case-by-case basis and allocate small power provisions to satisfy individual installation requirements:

— *Circulation corridors*: single socket outlets to be provided typically at 12 m maximum intervals within circulation areas for cleaning purposes.

— *Plantrooms*: single socket outlet typically to be provided for every 20 m^2 (approx.) for general maintenance purposes.

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— *Internal car parking*: single socket outlet to be provided for every 100 m^2 (approx.) for general maintenance purposes.

In office areas (typical minimum requirements only; final list to be provided as required considering usage, area and other project specifics):

— *Workstation (general)*: twin socket outlet to be provided for each workstation as a minimum with additional socket provided if necessary.

— *Meeting or conference tables*: twin socket outlet to be provided as a minimum, with one typically provided for every two chairs. (*Not*e: this is in addition to fixed power required for audio-visual equipment.)

— *Modular offices or meeting rooms*: desks serviced as above, plus additional socket outlets provided for general maintenance purposes (one single outlet as a minimum) and any other equipment likely to be installed (printers, audio-visual, etc.).

— *Open office (general)*: additional 10% (minimum) of total number of desk outlets to be provided for cleaning purposes and to serve other portable appliances such as printers, etc.

In residential applications (typical minimum requirements only; final list to be provided as required considering usage, area, quality level and other project specifics:

— *Kitchen*: four general socket outlets typically to be provided in addition to supplies dedicated to fixed kitchen equipment (e.g. fridge, freezer, washing machine, dishwasher, oven, hob, etc.).

— *Living room*: five socket outlets to be provided as a minimum (typical), with more provided as required to suit application.

— *Dining room, study, family rooms (where applicable)*: four socket outlets as a minimum, with more provided as required to suit application.

— *Bedroom*: four socket outlets as a minimum, with more provided as required.

— *Circulation hallway*: single socket outlet to typically be provided as a minimum.

— *Bathroom*: shaver socket outlets to be provided.

— *General*: all rooms to be equipped with at least one single socket as a minimum.

All small power outlets should typically be the three-pin flat shuttered type, rated at 13 amperes, in accordance with BS 1363 (BSI, 2016–2020).

Industrial socket outlets (400 V, 230 V) should also be provided to satisfy the power provision for specific installation types (e.g. workshops, plant rooms, industrial environments, etc.) and as required to satisfy final connection to certain specific equipment such as data or communications cabinet racks.

Fixed power supplies should be provided for all items of fixed equipment, in the form of fused connection units, double power switches, isolators and industrial socket outlets as appropriate.

Fixed power supplies should be considered for, but not necessarily be limited to, the following type of equipment:

— mechanical, public health and firefighting equipment

— communications, security and audio-visual equipment

— kitchen and catering equipment

— miscellaneous equipment, such as vending machines, water coolers, etc.

— other fixtures, furniture or electrical equipment

— equipment supplied or required by the end user.

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Designers should coordinate the location, rating and type of all fixed power outlets with other design disciplines, specialist installers, product suppliers and the end user.

Circuitry should be arranged with due consideration to equipment type, operational specifics and regulatory and authority compliance.

Ring mains should typically be considered for the general small power installation, with radial circuits typically considered for the fixed power installation and selected small power outlets. High-integrity earthing arrangements should be considered for socket outlets in commercial environments.

As part of the circuitry arrangements, designers should consider Residential and Commercial Buildings Electrical Installation Guidance for 400/230 V voltage system requirements for earth leakage protection. This should be provided in the form of individual residual-current device (RCD) circuit protection and/or grouped residual-current circuit breakers with overcurrent protection (RCBOs). Arrangements should be selected according to application and implemented with due consideration given to potential nuisance tripping and project type.

Accessories should typically be recessed in front of house areas, with surface-mounted types typically only allowed in non-visible areas (e.g. above ceiling voids) and back-of-house areas.

Accessory types and finishes should be selected to suit installation position, with front-of-house types being selected with due consideration to architecture and interior design arrangements and functions, and back-of-house types selected considering area type and usage. White plastic types should typically be considered for most general back-of-house applications (subject to architectural approval), with metal-clad types typically being considered for plant applications. Consideration should, however, be given to the presence of wet services or potential mechanical impact, with ingress-protected types and/or impact resistant-types used as appropriate.

9 Lighting

Lighting products should be selected to ensure the most appropriate design solutions are adopted for each individual area of all projects. In selecting the appropriate solutions, designers should consider aspects such as (but not limited to):

— project type

— area function, usage, and arrangement of furniture and equipment

— environmental considerations (i.e. internal or external, presence of wet services, etc.)

— area lighting requirements and design criteria

— architecture and interior design arrangements

— energy performance targets

— maintenance, including lamp replacement and cleaning

— installation restrictions and services coordination

— project budgetary constraints.

Lighting designs should offer a good quality of light throughout all areas, with consideration given not only to horizontal illuminance requirements (task illuminance) but also to vertical illuminance and uniformity and points of emphasis, such as architectural features, entrances, notional circulation routes, etc.

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In developing lighting calculations designers should apply relevant consideration to design factors, such as:

- lux levels
- uniformity
- surface mounting height
- working plane
- reflectance values (%)
- unified glare rating (UGR)
- colour rendering index
- ingress protection
- impact protection
- corrosion class
- explosion class
- maintenance factor that shall be designed with respect to:
  - room environment condition
  - room maintenance cycle
  - luminaire cleaning cycle
  - luminaire flux fraction
  - luminaire type
  - lamp type
  - luminaire maintenance factor/interval (LMF)
  - room surface maintenance factor/interval (RSMF)
  - lamp lumen maintenance factor (burning hours) (LLMF)
  - lamp survival factor (LSF).

Lighting designs should generally be developed considering the recommendations and design advice provided in Society of Light and Lighting (SLL) guidance.

Energy efficiency and consumption should be considered one of the prime drivers for luminaire selections, target lighting levels and control arrangements, with particular consideration given to the environmental building rating system for construction of new buildings in Saudi Arabia (Mostadam) requirements for each specific project. To reduce energy, designers should consider new luminaire and control technologies, efficient luminaire control gear (e.g. the use of electronic high frequency fluorescent ballasts in lieu of switch-start fluorescent ballasts), luminaire light output ratios, surface finishes, target light levels, etc.

Designers should offer appropriate consideration to luminaire maintenance including re-lamping, cleaning, product replacement, etc., as part of the final design solution. Particular attention should be given to areas such as atria, double height spaces, swimming pools, congested plant areas, external luminaires, etc.

Where luminaires are installed externally, they should be dustproof and waterproof, with suitable ingress protection ratings (typically IP65 as a minimum) and rated for continuous use at the high ambient temperatures and humidity levels prevalent locally (50 °C, 100% relative humidity).

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9.1 Lighting control

Designers should select the most appropriate means of lighting control for each area, considering:

- project type
- area usage
- functionality requirements
- presence of daylighting
- typical occupancy profile of area
- energy targets and efficiency
- end user and/or operator requirements.

Energy efficiency and energy savings should be considered a prime driver for the final lighting control selections, with automated controls (including the presence or absence and daylight detection) considered wherever possible. Where automated controls are not implemented, consideration should be given to time-based controls that ensure the lighting installation is not left energised unnecessarily when not in use.

The use of automated controls should not compromise the safety, functionality or practicality of an installation with alternative measures considered as required. For instance, the use of automated controls could present issues in the following sample areas and therefore should not be considered:

- *Plantrooms*: automated controls in these areas could be a potential safety risk. Occupancy sensors could turn off lights during critical maintenance; ductwork, pipework and general services reticulation may make sensors unviable in these areas. Local occupant controls, via traditional on/off switches, would generally be more suitable.
- *Meeting rooms and conference areas*: automated controls in these areas would lead to functionality issues as the lighting installation is likely to be layered, with multiple arrangements required to suit usage. Scene-setting controls are likely to be more suitable in such areas.
- *Residential accommodation*: automated controls in these areas would not always be suitable, with occupants requiring localised control to suit requirements. Traditional on/off switches and/or dimmers would generally be more suitable in such areas.

The use of traditional manual switches should generally be limited to areas where automated controls are not deemed appropriate — such as electrical rooms, residential accommodation, etc. — or where used in conjunction with automated controls (e.g. manual switches used to energise luminaires, with automated presence-detection controls used to reduce lighting levels and ultimately de-energise luminaires when areas are not in use).

In areas where manual control is required over layers of lighting and the lighting levels (e.g. meeting rooms, reception areas, classrooms, etc.), and where lighting requirements will vary depending on specific uses at different points in time, dimming systems should generally be considered. In such cases, consideration should also be given to links and interfaces to any associated audio-visual system provisions where applicable.

Consideration should also be given to the control of lighting via centralised provisions (i.e. through a building management system where provided) so that operations can be scheduled and monitored accordingly. This should particularly concern common and circulation areas. Designers should ensure that all control provisions are suitable for each area, and that the final control provisions are fully compatible with the respective luminaires. All control provisions should allow flexibility and future extendibility of the installation.

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9.2 Emergency lighting

Emergency lighting should be considered for all projects in accordance with requirements of:

— SBC-801: *Saudi Fire Protection Code* (SBCNC, 2018c)

— NFPA 70: *National Electrical Code* (NFPA, 2020a)

— NFPA 101: *Life Safety Code* (NFPA, 2018).

— BS 5266-1: *Emergency lighting. Code of practice for the emergency lighting of premises* (BSI, 2016)

— BS EN 1838: *Lighting applications. Emergency lighting* (BSI, 2013).

Emergency lighting should be designed to automatically provide sufficient illumination to allow safe exit from any building in the event of any general or localised failure of the normal electrical supply.

9.2.1 Design criteria

Emergency lighting design proposals should be developed in accordance with the lighting levels stipulated within all relevant National Fire Protection Association (NFPA) documentation and as required to achieve Saudi Civil Defence approvals. Designers should note that the achievement of Saudi Civil Defence approval does not negate the requirement to comply with NFPA requirements. Both sets of requirements should be considered, with the most onerous adopted throughout designs.

The below list provides a summary of target emergency lighting levels based on NFPA requirements applicable at the time of developing this document. This information is provided for reference purposes only and designers should refer to the latest NFPA documentation for further information.

Emergency escape lighting to be arranged along the path of egress to provide initial illumination of:

— not less than an average of 10.8 lux at finished floor level

— not less than a minimum of 10.8 lux at finished floor level (high risk areas)

— not less than a minimum of 1.1 lux at finished floor level

— not less than a minimum 0.5 lux at the floor level of the circulation of any open area (anti-panic lighting).

The positions of emergency escape lighting to be considered shall include the following:

— at each exit door intended to be used in an emergency

— near stairs so that each flight of stairs receives direct light

— near any other change in level

— mandatory emergency exits and safety signs

— at each change of direction

— at each intersection of corridors

— outside and near to each final exit

— near each first aid post

— near each item of fire fighting equipment

— equipment and call point.

Emergency lighting design should ensure that a maximum to minimum illumination uniformity ratio of 40:1 is not exceeded.

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Saudi Civil Defence requires emergency lighting back-up durations to be minimum 90 minutes, the same as the NFPA requirement.

In areas of high risk the maintained illuminance on the reference plane shall be not less than 10 % of the required maintained illuminance for that task; however, it shall be not less than 15 lux.

The temperatures in the areas where the batteries are installed should be considered in achieving the required lighting levels for the defined back-up durations. This should particularly concern self-contained emergency lighting solutions where luminaires, complete with batteries, may be installed within untreated, non-air conditioned areas where the applied temperatures may adversely affect battery durations.

Designers should seek manufacturer advice and confirmations in such instances to ensure the proposals achieve the requirements defined herein and those governed by the associated codes and regulations.

9.2.2 Emergency luminaire types

Designers should offer consideration to emergency luminaire types on a case-by-case basis, considering:

- project type and budget
- area type and function
- aesthetics of project and of area, and architectural and interior design arrangements
- ceiling proposals
- normal/general lighting proposals
- emergency lighting design requirements.

Exit luminaires should be selected to suit applications and be appropriately positioned to clearly denote the exit route and/or final exit doors.

Designers should consider exit sign size as part of the design and ensure viewing distances are arranged in compliance with the aforementioned codes, regulations and authority requirements.

9.2.3 System types

Emergency lighting system type should be dependent on project type, the size, number and load of luminaires, and the requirements of local civil defence, associated codes and the end user.

Central battery emergency lighting systems should be used for all high-rise projects and considered for all medium- to large-scale projects. Self-contained emergency lighting proposals, incorporating battery packs, inverters, etc. within each emergency luminaire should be considered only on small projects where maintenance is more manageable.

Designers should offer consideration of system types at the start of each project considering various factors as part of the system appraisal.

This should include but not necessarily be limited to:

- operational management
- maintenance and system diagnostics
- testing
- capital and operational costs

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— code, regulation and authority requirements.

All emergency luminaires should be provided with testing facilities. Emergency lighting shall be inspected and tested in accordance with SBC-801 sections 604.6.1 to 604.6.2.1 inclusive (SBCNC, 2018c). Activation test shall be completed monthly, records of tests shall be maintained (include locations of emergency lighting tested (passed/failed), date of the test, and person completed the test), power test record (for battery-powered emergency lighting, a power test of the emergency lighting equipment shall be completed annually for not less than 90 minutes and shall remain sufficiently illuminated for the duration of the test).

Automatic programmed central testing facilities complete with luminaire diagnostics should be considered, via central battery system (where provided) or central test facility. Manual testing provisions will be accepted only for selected applications and should be limited to all but the smallest of projects.

Central testing facilities and central battery systems should, where provided, be connected to a building management system and/or master control PC enabling daily reports and logs to be generated, the programming of scheduled tests and the operation of the system in selected conditions (e.g. fire alarm activation).

All wiring associated with the emergency wiring installation (from batteries to luminaire) should be carried out with fire-resistant cabling meeting NFPA, IEC, Saudi Civil Defence and SBC-801 requirements. Wiring and associated equipment should be arranged so that associated emergency luminaires operate in the event of any local circuit failure or mains supply failure.

10 Aviation obstruction lighting

Designers should consider the need for aviation obstruction lighting (AOL) at the start of each project. System designs should be subjected to an independent aeronautical study that assesses the location of the building or plot in relation to the flight path and highlight the following as required:

— AOL requirement

— type of AOL equipment required

— quantity of luminaires required

— AOL sequencing, operations and controls.

The aeronautical study and subsequent system design (where appropriate) should be undertaken in accordance with the requirements and standards of GACA (General Authority of Civil Aviation) and the International Civil Aviation Organisation (ICAO). Where AOL systems are deemed required, they should provide sufficient light to the external façades of buildings to highlight the building to aviation traffic being routed through the vicinity. Consideration should be given in the placement of AOL luminaires to ensure that no discomfort glare is experienced by occupants of surrounding buildings.

All AOL systems should be supplied via a dedicated uninterruptible power supply (UPS) located in the same location as the system controller. The UPS should be sized according to the number of luminaires installed, with battery autonomy provided in accordance with the requirements of the aeronautical study.

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11 Lightning protection

11.1 General

Designers should carry out a lightning protection assessment on all projects in accordance with BS EN 62305/IEC 62305 (BSI, 2011–2012; IEC, 2010a) and/or NFPA 780 (NFPA, 2020b). This assessment should determine if lightning protection is required, and if so, what level of classification (lightning protection level (LPL)) the design needs to achieve. Designers should verify and validate assessments with specialist suppliers as part of the design process to ensure final provisions are arranged accordingly.

Lightning protection arrangements (where required) should be designed in accordance with IEC 62305 with consideration also given to BS 7430 (BSI, 2011/2015), BS 7671 (BSI, 2018/2020).

Protection systems should comprise the following main elements as a minimum:

— air termination network

— down conductors

— earth termination network, including test points.

Material used for lightning protection should conform to the aforementioned requirements and be arranged so that the impact on buildings is minimised as far as possible. Consideration should be given to the use of natural components of the building, such as rebars within structural perimeter columns, to eliminate exposed tapes on building façades. Where such measures are employed the designer should ensure that specific structural arrangements are incorporated into the design and that the arrangements are acceptable to the Saudi Electricity Company (SEC).

The arrangement of lightning protection components (e.g. spacing of down conductors) should be directly related to the assessed classification of lightning protection for the specific project and the structural grid arrangements.

All metallic projections and plant or equipment on or above the main surface of the roof structure should be bonded to the lightning protection system or protected via an air terminal arrangement. All bonds associated should be suitably treated to suit the local environmental conditions. All down conductors should terminate at dedicated testing points prior to final connection to earth.

These should allow regular routine testing in accordance with the associated regulations. Final earth connection should typically be arranged in the form of earth pits complete with earth electrode rods, dedicated to each down conductor. In certain cases, the use of structure pile foundations (where applicable) may be considered to avoid the installation of separate pits and minimise slab penetrations. Where such arrangements are adopted the designer should ensure that appropriate measures are incorporated into the structural design and that the arrangements are acceptable to SEC.

12 Earthing and bonding

Designers shall carry out earthing calculations by using accredited software of the system manufacturer for all each project based on the prospective fault current of the power supply and soil resistivity in accordance with relevant soil report.

Project designers should provide separate dedicated earthing provisions (earth rod electrodes) for each project. Earth connections, together with all internal earthing and bonding arrangements, should be designed in accordance with the following:

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- SBC-401 (SBCNC, 2018b) regulations
- IEC 61936-1 (IEC, 2010b)
- BS 7671 (BSI, 2018/2020)
- ANSI/BICSI 002-2019 (BICSI, 2019) for data centres
- TIA-942-B (TIA, 2017) for data centres
- TIA-607-D (TIA, 2019) for telecommunications.

Earthing systems should comprise earth electrodes, main earthing cables between electrodes and earth bars, main earth bars, earth continuity conductors (ECCs) for all outgoing sub-main and final circuits and equipotential bonding of all metalwork and exposed conductive parts.

Separate earth electrodes and dedicated earth bars should typically be provided for the following systems and/or rooms:

- transformer neutral
- transformer body
- MV room
- LV switch panels
- generator room
- generator neutral
- telecoms provider intake rooms
- project-specific requirements, such as hospital operating rooms.

A separate earthing system (clean earth) will be provided for telecommunication, electronic and data systems.

A minimum of two earth electrodes should typically be provided for each main system/sub-system, with two connections provided to each earth bar to enable one to be disconnected for test purposes while the other is in circuit. All earth bars should be provided with two isolation points for similar purposes.

The main earth electrode resistance should not exceed the maximum level dictated by the Saudi Electricity Company (SEC). Electrodes should be spaced at least equal to their length, with six metres provided between each as a minimum.

Earth pits should be specified considering potential weight applied, with specific consideration given to pits located in areas subject to vehicular access.

Dedicated earth continuity conductors (ECCs) should be provided for the earth bars for each circuit. Designers should calculate all ECCs in compliance with the regulations and should note that SBC-401 regulations define the minimum cable sizes required only.

Equipotential bonding should be provided to connect all items of extraneous metal work to the general mass of earth to reduce potential shock hazards. This should include (but not be limited to) the building structure, sinks, raised access floor pedestals, MEP plant (i.e., HVAC systems, domestic water pumping systems, sprinkler pipework, gas pipework, etc.) All electrical switch rooms and sub-main rooms and risers should be provided with a copper earth bar for equipotential bonding purposes.

All earthing and bonding requirements should be in accordance with SBC-401 regulations and requirements.

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Appendix 1: Saudi Electricity Company documents

A1.1 MV and LV design documents

Table A1.1 Distribution Planning Standards (DPS)

Title	Code
Distribution Planning Standard	DPS
Estimation of Customer Load Guideline	DPS 01
Design Guideline of Underground Low Voltage Network to Supply Customers	DPS 02
Multi Source Two or One Voltage Supplies, Including More Than One MVA	DPS 06
Design Guideline of Residential and Commercial to Supply Customers (Arabic)	DPS 09
Design Guidelines for Residential (Arabic)	DPS 10

Table A1.2 Distribution Material Specifications (SDMS)

Title	Code
General Requirements for All Equipment/Material	01-SDMS-01
13.8 and 33 kV Overhead Line Conductors (ACSR/AW Type)	10-SDMS-01
Bare Copper Conductors	10-SDMS-02
Low Voltage Power and Control Cables	11-SDMS-01
LV Overhead Line Conductor Type Quadruplex	11-SDMS-02
XLPE Insulated Power Cables for Rated Voltages from 15 kV up to 36 kV	11-SDMS-03
Specification for Aluminum Unarmored XLPE/Lldpe Insulated Power Cables for Rated Voltages from 15 kV up to 36 kV	11-SDMS-04
Cable Joints, Terminations and Accessories up to 36 kV	12-SDMS-01
Lugs and Connectors for MV/LV Distribution System	12-SDMS-02
Specification for Underground Warning Tapes	12-SDMS-03
Specification for Cable Clamps For Power Cables	12-SDMS-04
Specification for Optical Fiber Cable Splicing	12-SDMS-05
Overhead Line Porcelain Insulators	15-SDMS-01
Overhead Line Polymeric Insulators	15-SDMS-02
Specification for Octagonal Steel Poles	20-SDMD-01
Octagonal Steel Poles for Distribution Lines	20-SDMS-01
Specification for Overhead Line Accessories	20-SDMS-02
Over Headline Accessories	20-SDMS-02
Prestressed Spun Concrete Pole for Distribution System	20-SDMS-03
Specification for Duct, HDPE Main Mini and Micro-Ducts	23-SDMS-01
Load Break Switch	30-SDMS-01
Disconnect and Bypass-Disconnect Switches	30-SDMS-02
Automatic Transfer Switch For MV Network	30-SDMS-03
LBS SF6 — GAS	30-SDMS-04
Low Voltage Distribution Panel	31-SDMS-01
LV Distribution Pillar (Mini-Pillar)	31-SDMS-02
Low Voltage Cabinet for Pole Mounted Transformer	31-SDMS-03
Specifications for 400/231 V Low Voltage Cabinet with Aluminum Busbars for Pole Mounted Transformers	31-SDMS-03A
Low Voltage Distribution Panel without Outgoing Circuit Breakers	31-SDMS-04
400/230 Volts Low Voltage Distribution Panel	31-SDMS-05

Table continues

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Table A1.2 Distribution Material Specifications (SDMS) (*continued*)

Title	Code
400/230 Volts Low Voltage Distribution Panel without Outgoing Circuit Breakers	31-SDMS-06
Low Voltage Distribution Panel with Aluminum Bus Bars	31-SDMS-07
Specifications for Low Voltage Distribution Panel with Aluminum Busbar, Main Circuit Breaker and Outgoing MCCBs	31-SDMS-08
Specifications for Auxiliary AC/DC Panels for Primary Distribution Substations	31-SDMS-10
Terminal Blocks for Primary Distribution Substations	31-SDMS-11
Specifications for Relay And Control Panels for Primary Distribution Substations	31-SDMS-12
Automatic Transfer Switch for Low Voltage Network	31-SDMS-13
Non-Extensible SF6 Insulated Ring Main Unit, 17.5 kV	32-SDMS-01
MV Metered Ring Main Unit, up to 36 kV	32-SDMS-02
Extensible SF6 Insulated Ring Main Unit, 17.5 kV	32-SDMS-03
Motorized Non-Extensible SF6 Insulated Ring Main Unit, 17.5 kV	32-SDMS-04
630 Amperes MV Metered Ring Main up to 36 kV	32-SDMS-05
630 Amperes MV Metered Ring Main up to 38 kV	32-SDMS-06
Non-Extensible SF6 Insulated Ring Main Unit, 36 kV	32-SDMS-07
Metal Clad Switchgear for Primary Distribution	32-SDMS-08
Specifications for Gas Insulated Metal Clad MV Switchgear (11, 13.8, 33 and 34.5 kV)	32-SDMS-09
Specifications for Gas Insulated Metal Clad 69 kV Switchgear	32-SDMS-10
Specifications for Automated RMU up to 36 kV	32-SDMS-11
MV Auto Reclosers up to 36 kV	33-SDMS-01
MV Automatic Line Sectionlizer up to 36 kV	33-SDMS-02
MV Dropout Fuse Cut Outs	34-SDMS-01
MV and LV Fuse Links	34-SDMS-02
Metal Oxide Surge Arresters	35-SDMS-01
Antenna Measurement for Medium Voltage 33 kV and 13.8 kV	36-SDMS-01
Low Voltage Molded Case Circuit Breakers for Service Connections	37-SDMS-01
Molded Case Circuit Breakers for Low Voltage Distribution Panels 400 Amps	37-SDMS-02
Molded Case Circuit Breakers for Low Voltage PMT Cabinet 200 Amps	37-SDMS-03
Interface Low Voltage Main Circuit Breakers	37-SDMS-04
Fault Indicators for Underground Distribution Networks	38-SDMS-01
Fault Indicators for Overhead Distribution Networks	38-SDMS-02
LV Digital Panel Meters	38-SDMS-03
Specification For Annunciator Systems for Primary Distribution Substation	38-SDMS-04
Smart OHL Fault Indicators	38-SDMS-05
Specifications for Smart Earth Fault Indicators for Underground Distribution Networks	38-SDMS-06
Bottom Connected Kilowatt-Hour Meter	40-SDMS-01
Electronic Revenue CT and CT/VT Meters	40-SDMS-02A
Supplementary (SEC DLMS/COSEM OBIS Code Specification) for Electronic Revenue CT And CT/VT Meters	40-SDMS-02A
Electronic Revenue Whole Current Meters	40-SDMS-02B
Supplementary (SEC DLMS/COSEM OBIS Code Specification) for Electronic Revenue WC Meter	40-SDMS-02B
Fiberglass Reinforced Polyester Meter Boxes	42-SDMS-01
Fiberglass Reinforced Low Voltage Service Cabinet for 800 A Bulk Customers	42-SDMS-02

Table continues

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Table A1.2 Distribution Material Specifications (SDMS) (*continued*)

Title	Code
Inside Room Fiberglass Reinforced Polyester Meter Boxes	42-SDMS-03
MV Shunt Power Capacitor Bank up to 36 kV	43-SDMS-01
MV Step Voltage Regulators up to 36 kV	43-SDMS-02
Power Factor Correction for Low Voltage Power Systems	43-SDMS-03
Specifications for Shunt Capacitor Bank 13.8 kV Through 69 kV for Primary Distribution Substation	43-SDMS-04
Vented Nickel-Cadmium Stationary Battery Bank for Primary Distribution Substations	46-SDMS-01
Battery Chargers for Primary Distribution Substations	46-SDMS-02
Specification for Current Transformers Rated up to 36 kV	50-SDMS-01
Primary Substation Current Transformers 11 kV Through 69 kV	50-SDMS-02
Primary Substation Potential Transformers 11 kV through 69 kV	50-SDMS-03
Distribution Transformers up to 36 kV	51-SDMS-01
Distribution Transformers up to 36 kV 400/230 Volts	51-SDMS-02
Distribution Transformers up to 36 kV 400/230 Volts With Aluminum Windings	51-SDMS-03
Distribution Transformers up to 36 kV With Aluminum Windings for Dual Secondary 400/230 V and 230/133 V	51-SDMS-04
Power Transformers for Primary Distribution System	53-SDMS-01
13.8 kV Unit Substations	56-SDMS-01
13.8 kV Package Substations	56-SDMS-02
13.8 kV Unit Substations 400/230 V	56-SDMS-03
Specifications for 13.8 kV Package Substations 400/230 Volts	56-SDMS-04
Two Trailer Mounted Mobile Substations, 33/13.8 kV	56-SDMS-05
Single Trailer Mounted Mobile Substations, 33/13.8 kV	56-SDMS-06
Unit Substations up to 36 kV Aluminum Transformer, Aluminum Bus Bars LVDP	56-SDMS-07
MV/LV Mobile Substation	56-SDMS-08
Unit Substations up to 36 kV Aluminum Transformer, Aluminum Bus Bars LVDP with Main Circuit Breaker and Outgoing MCCBs	56-SDMS-09
Neutral Grounding Resistor 11 kV through 34.5 kV for Primary Distribution Substations	57-SDMS-01
Air Core Series Reactors 13.8 kV through 69 kV for Primary Distribution Substations	58-SDMS-01
Monogram	SEC-01-01
Insulating Oil for Transformers and Switchgear	SEC-02-01
Pad Locks and Cylindrical Locks	SEC-02-02
Specification for Single-Piece Metallic High-Security Cable Seal	SEC-03-01

Table A1.3 Distribution Construction Standards (SDCS)

Title	Code
Concrete Foundations	SDCS Part 1
Installation of kWh Meter Boxes On Front Consumer Walls and Side Insets	SDCS Part 2
Installation of Recessed Type kWh Meter Boxes into Consumer's Wall	SDCS Part 3
Installation of Distribution Pillars	SDCS Part 4
Installation of Fiberglass Reinforced Low Voltage Service Cabinet for 800 A Consumers	SDCS Part 5
Installation of kWh Meter Boxes Inside Meter Rooms	SDCS Part 7
Protective Barriers	SDCS Part 8

Table continues

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Table A1.3 Distribution Construction Standards (SDCS) (*continued*)

Title	Code
Installation of 13.8 and 33 kV Single Transformer Room Substation	SDCS Part 9
13.8 and 33 kV Metered Ring Main Unit SF6 Room Substations	SDCS Part 10
Security Fence For Unit Substations And RMUs	SDCS Part 11
Unit Substation + RMU Layout	SDCS Part 12
Roof and Door Canopy Waterproofing	SDCS Part 13
Outdoor Metering Kiosk	SDCS Part 14
Single Meter Box for Temporary Project	SDCS Part 15
Manhole	SDCS Part 16
Cable Trenches	SDCS Part 17

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